



K.R. Mangalam University
School of Engineering & Technology

Fundamentals Of Java Programming Lab
(ENCA203) Assignment 4
Student Management System

Submitted by:

Name: Shivam Gaur

Roll no. 2401201016

Class: BCA (h) (AI & DS)

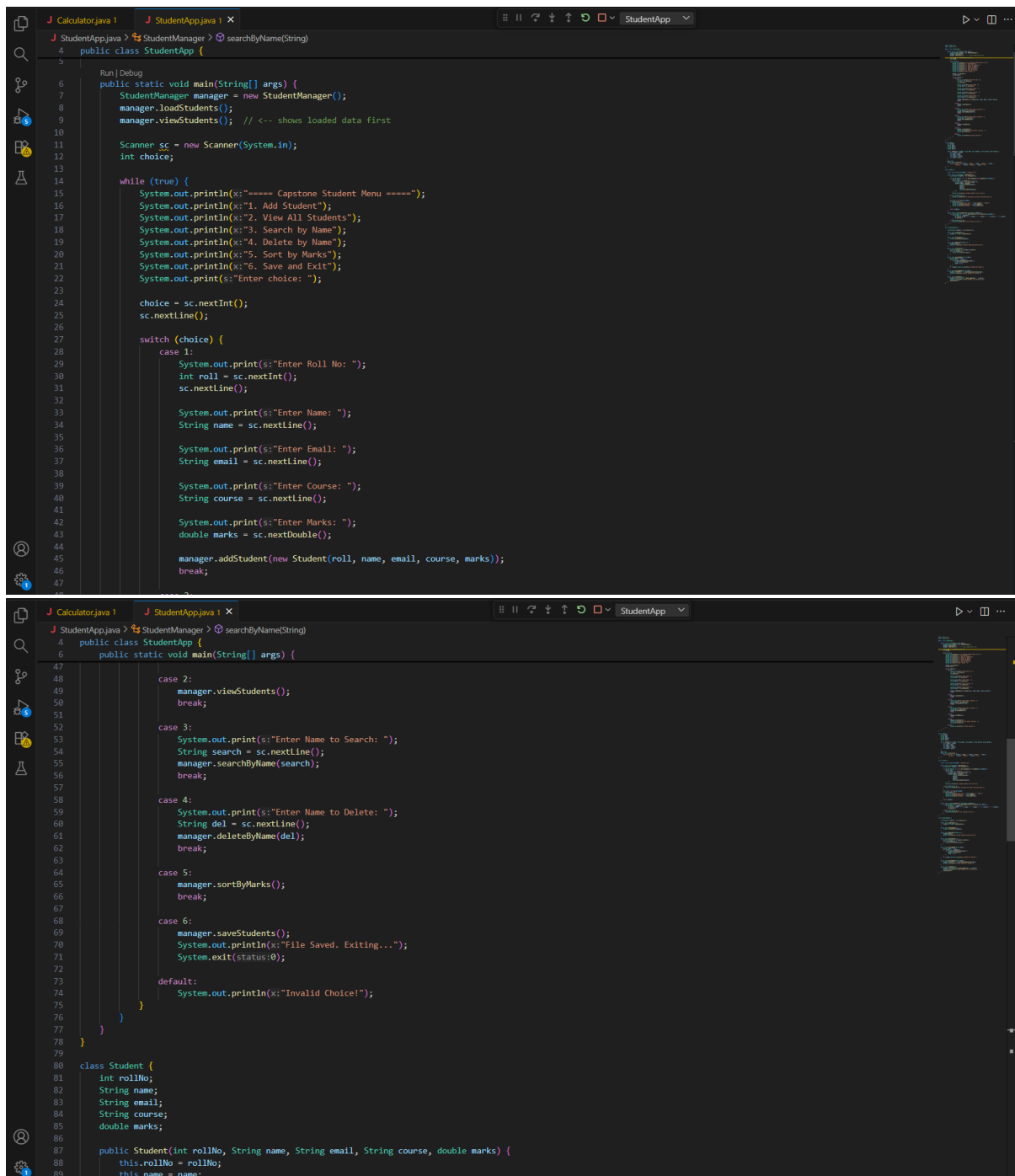
Submitted to:

Dr. Manish Kumar

GitHub Repository:

https://github.com/shivamgaur1255-sketch/java_assignments-

Codes:

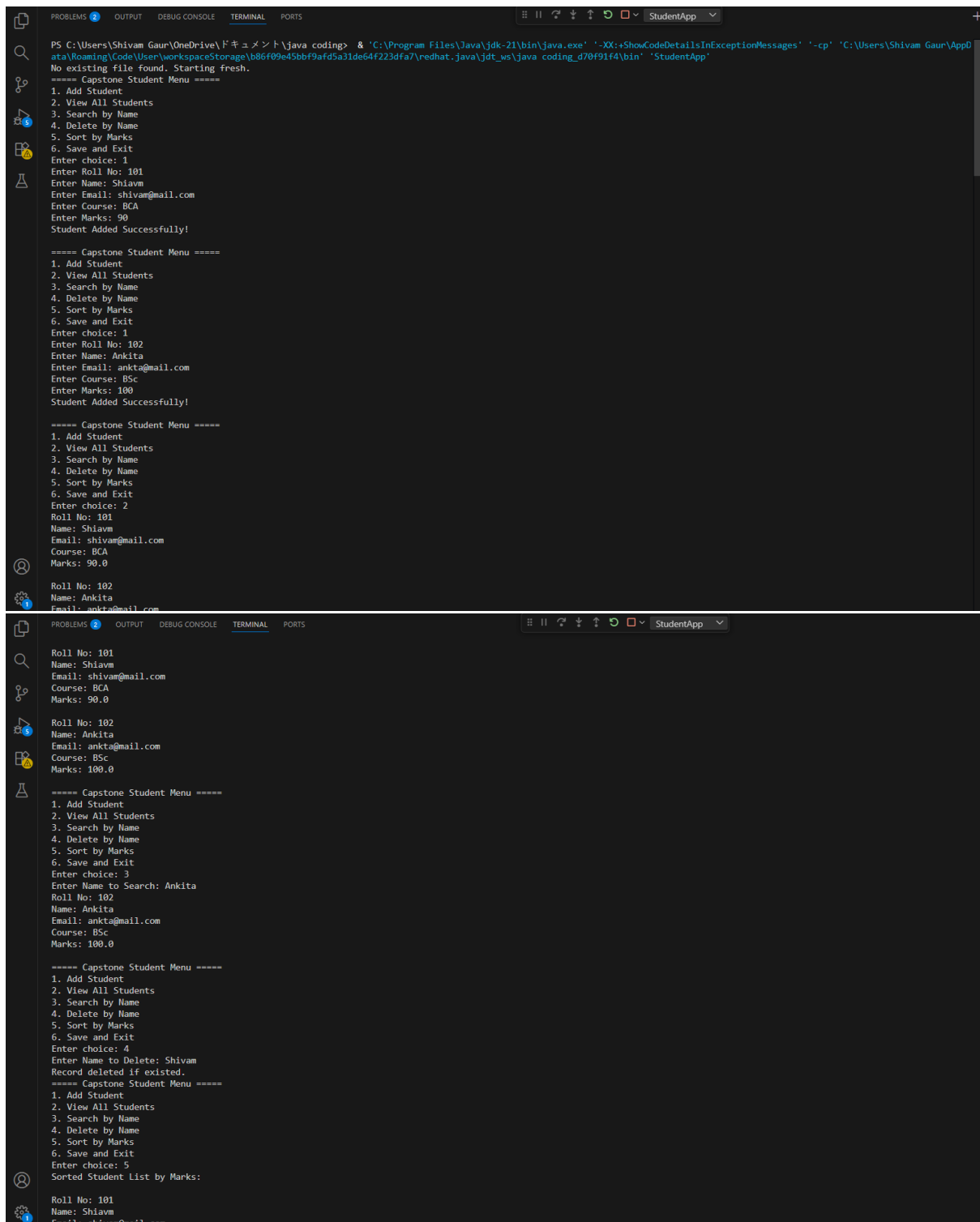


```
4 public class StudentApp {
5
6     public static void main(String[] args) {
7         StudentManager manager = new StudentManager();
8         manager.loadStudents();
9         manager.viewStudents(); // <-- shows loaded data first
10
11         Scanner sc = new Scanner(System.in);
12         int choice;
13
14         while (true) {
15             System.out.println(x:"==== Capstone Student Menu =====");
16             System.out.println(x:"1. Add Student");
17             System.out.println(x:"2. View All Students");
18             System.out.println(x:"3. Search by Name");
19             System.out.println(x:"4. Delete by Name");
20             System.out.println(x:"5. Sort by Marks");
21             System.out.println(x:"6. Save and Exit");
22             System.out.print(s:"Enter choice: ");
23
24             choice = sc.nextInt();
25             sc.nextLine();
26
27             switch (choice) {
28                 case 1:
29                     System.out.print(s:"Enter Roll No: ");
30                     int roll = sc.nextInt();
31                     sc.nextLine();
32
33                     System.out.print(s:"Enter Name: ");
34                     String name = sc.nextLine();
35
36                     System.out.print(s:"Enter Email: ");
37                     String email = sc.nextLine();
38
39                     System.out.print(s:"Enter Course: ");
40                     String course = sc.nextLine();
41
42                     System.out.print(s:"Enter Marks: ");
43                     double marks = sc.nextDouble();
44
45                     manager.addStudent(new Student(roll, name, email, course, marks));
46                     break;
47
48                 case 2:
49                     manager.viewStudents();
50                     break;
51
52                 case 3:
53                     System.out.print(s:"Enter Name to Search: ");
54                     String search = sc.nextLine();
55                     manager.searchByName(search);
56                     break;
57
58                 case 4:
59                     System.out.print(s:"Enter Name to Delete: ");
60                     String del = sc.nextLine();
61                     manager.deleteByName(del);
62                     break;
63
64                 case 5:
65                     manager.sortByMarks();
66                     break;
67
68                 case 6:
69                     manager.saveStudents();
70                     System.out.println(x:"File Saved. Exiting...");
71                     System.exit(status:0);
72
73                 default:
74                     System.out.println(x:"Invalid Choice!");
75             }
76         }
77     }
78 }
79
80 class Student {
81     int rollNo;
82     String name;
83     String email;
84     String course;
85     double marks;
86
87     public Student(int rollNo, String name, String email, String course, double marks) {
88         this.rollNo = rollNo;
89         this.name = name;
90     }
91 }
```

```
J Calculator.java | J StudentApp.java | X
J StudentApp.java | StudentManager | searchBy Name(String)
80 class Student {
81     public Student(int rollNo, String name, String email, String course, double marks) {
82         this.rollNo = rollNo;
83         this.name = name;
84         this.email = email;
85         this.course = course;
86         this.marks = marks;
87     }
88
89     @Override
90     public String toString() {
91         return "Roll No: " + rollNo + "\nName: " + name + "\nEmail: " + email +
92             "\nCourse: " + course + "\nMarks: " + marks + "\n";
93     }
94 }
95
96 class FileUtil {
97     static final String FILE_NAME = "students.txt";
98
99     public static List<Student> readFromFile() {
100         List<Student> students = new ArrayList<>();
101
102         try (BufferedReader br = new BufferedReader(new FileReader(FILE_NAME))) {
103             String line;
104             while ((line = br.readLine()) != null) {
105                 String[] data = line.split(",");
106                 students.add(new Student(
107                     Integer.parseInt(data[0]),
108                     data[1],
109                     data[2],
110                     data[3],
111                     Double.parseDouble(data[4])
112                 ));
113             }
114             System.out.println("Loaded students from file:\n");
115         } catch (IOException e) {
116             System.out.println("No existing file found. Starting fresh.");
117         }
118
119         File file = new File(FILE_NAME);
120         if (file.exists()) {
121             System.out.println("File Size: " + file.length() + " bytes");
122         }
123     }
124
125     public static void writeToFile(List<Student> students) {
126         try (BufferedWriter bw = new BufferedWriter(new FileWriter(FILE_NAME))) {
127             for (Student s : students) {
128                 bw.write(s.rollNo + "," + s.name + "," + s.email + "," + s.course + "," + s.marks);
129                 bw.newLine();
130             }
131         } catch (IOException e) {
132             System.out.println("Error writing to file.");
133         }
134     }
135 }
136
137 class StudentManager {
138     List<Student> students = new ArrayList<>();
139
140     public void loadStudents() {
141         students = FileUtil.readFromFile();
142     }
143
144     public void saveStudents() {
145         FileUtil.writeToFile(students);
146     }
147
148     public void addStudent(Student s) {
149         students.add(s);
150         System.out.println("Student Added Successfully!\n");
151     }
152
153     public void viewStudents() {
154         Iterator<Student> it = students.iterator();
155         while (it.hasNext()) {
156             System.out.println(it.next());
157         }
158     }
159 }
160
161 }
```

```
J Calculator.java | J StudentApp.java | StudentApp
J StudentApp.java | StudentManager | searchByName(String)
149 class StudentManager {
151     public void addStudent(Student s) {
154     }
155
156     public void viewStudents() {
157         Iterator<Student> it = students.iterator();
158         while (it.hasNext()) {
159             System.out.println(it.next());
160         }
161     }
162
163     public void searchByName(String name) {
164         boolean found = false;
165         for (Student s : students) {
166             if (s.name.equalsIgnoreCase(name)) {
167                 System.out.println(s);
168                 found = true;
169             }
170         }
171         if (!found) System.out.println("Student Not Found!");
172     }
173
174     public void deleteByName(String name) {
175         students.removeIf(s -> s.name.equalsIgnoreCase(name));
176         System.out.println("Record deleted if existed.");
177     }
178
179     public void sortByMarks() {
180         students.sort(Comparator.comparingDouble(s -> s.marks));
181         System.out.println("Sorted Student List by Marks:\n");
182         viewStudents();
183     }
184 }
185
186
187
188
189
190
191
192
193
194
195
```

Output:



```
PS C:\Users\Shivam Gaur\OneDrive\ドキュメント\java coding> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Shivam Gaur\AppData\Roaming\Code\User\workspaceStorage\b86f09e45bbf9afd5a31de64f223dfa7\redhat.java\jdt_ws\java_coding_d70f91f4\bin' 'StudentApp'
No existing file found. Starting fresh.
===== Capstone Student Menu =====
1. Add Student
2. View All Students
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
Enter choice: 1
Enter Roll No: 101
Enter Name: Shivam
Enter Email: shivam@mail.com
Enter Course: BCA
Enter Marks: 90
Student Added Successfully!

===== Capstone Student Menu =====
1. Add Student
2. View All Students
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
Enter choice: 1
Enter Roll No: 102
Enter Name: Ankita
Enter Email: ankita@mail.com
Enter Course: BSc
Enter Marks: 100
Student Added Successfully!

===== Capstone Student Menu =====
1. Add Student
2. View All Students
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
Enter choice: 2
Roll No: 101
Name: Shivam
Email: shivam@mail.com
Course: BCA
Marks: 90.0

Roll No: 102
Name: Ankita
Email: ankita@mail.com

Roll No: 101
Name: Shivam
Email: shivam@mail.com
Course: BCA
Marks: 90.0

Roll No: 102
Name: Ankita
Email: ankita@mail.com
Course: BSc
Marks: 100.0

===== Capstone Student Menu =====
1. Add Student
2. View All Students
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
Enter choice: 3
Enter Name to Search: Ankita
Roll No: 102
Name: Ankita
Email: ankita@mail.com
Course: BSc
Marks: 100.0

===== Capstone Student Menu =====
1. Add Student
2. View All Students
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
Enter choice: 4
Enter Name to Delete: Shivam
Record deleted if existed.
===== Capstone Student Menu =====
1. Add Student
2. View All Students
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
Enter choice: 5
Sorted Student List by Marks:

Roll No: 101
Name: Shivam
Email: shivam@mail.com
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
StudentApp

Course: BCA
Marks: 90.0

Roll No: 102
Name: Ankita
Email: ankta@mail.com
Course: BSc
Marks: 100.0

===== Capstone Student Menu =====
1. Add Student
2. View All Students
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
Enter choice: 5
Sorted Student List by Marks:

Roll No: 101
Name: Shivam
Email: shivam@mail.com
Course: BCA
Marks: 90.0

Email: shivam@mail.com
Course: BCA
Marks: 90.0

Course: BCA
Marks: 90.0

Marks: 90.0

Roll No: 102
Name: Ankita
Email: ankta@mail.com
Roll No: 102
Name: Ankita
Email: ankta@mail.com
Email: ankta@mail.com
Course: BSc
Marks: 100.0
Course: BSc
Marks: 100.0
Marks: 100.0

===== Capstone Student Menu =====
===== Capstone Student Menu =====
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
StudentApp

Roll No: 102
Name: Ankita
Email: ankta@mail.com
Roll No: 102
Name: Ankita
Email: ankta@mail.com
Email: ankta@mail.com
Course: BSc
Marks: 100.0
Course: BSc
Marks: 100.0
Marks: 100.0

===== Capstone Student Menu =====
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1. Add Student
2. View All Students
1. Add Student
2. View All Students
2. View All Students
3. Search by Name
4. Delete by Name
3. Search by Name
4. Delete by Name
5. Sort by Marks
6. Save and Exit
4. Delete by Name
5. Sort by Marks
6. Save and Exit
5. Sort by Marks
6. Save and Exit
Enter choice: 6
Enter choice: 6
File Saved. Exiting...
PS C:\Users\Shivam Gaur\OneDrive\ドキュメント\java coding>
```